Wenjie Guan

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Education

Cornell University – Ph.D. in Statistics Aug. 2024 – May. 2028 (expected)
GPA: 4.12. Advisor: Jelena Bradic. *Ithaca, NY*

University of Washington – M.S. in Statistics Sept. 2022 – Jun. 2024
GPA: 3.95/4.0 *Seattle, WA*

Fudan University – B.S. in Mathematics and Applied Math Sept. 2018 – Jun. 2022
GPA: 3.70/4.0 (with major GPA 3.88) *Shanghai, China*

Research Interests

Machine Learning Theory, Causal Inference and Robustness, Natural Language Processing, High-dimensional Statistics, Statistical Methodologies.

Work Experience

Handshake MoVE program Jul. 2025 – Feb. 2026
Math Fellow and Reviewer – part time *Remote*

- My duty is to write graduate-level math and statistics prompts that LLM models fail to answer or prove correctly, evaluate their reasoning capabilities and explain where they fail, and provide the step-by-step solutions to the prompts. Also I review the prompts provided by fellows and move the good ones to production.

Publications and Working Papers

W. Guan, J. Bradic. When Symbol Names Should Not Matter: A Logistic Theory of Fresh-Symbol Classification, In *the Conference on Neural Information Processing Systems (NeurIPS)*, 2026. Website, ArXiv.

W. Guan, J. Ma. Network Discovery and Clustering with Block-structured Gaussian Graphical Model, revision in progress at *Journal of Computational and Graphical Statistics*, 2026.

W. Guan, Y. Zhang, D. Zhou, T. Cai, A. Giessing. Efficient Inference on High-Dimensional Logistic Regression under Class Imbalance, under preparation.

Research Experience

Asymptotics for Online Double Machine Learning Sept 2026 – Now
Advisor: Jelena Bradic and Yang Ning, professors at Cornell University *Ithaca, NY*

Generalization of Transformers on Classification Tasks Oct 2025 – May 2026
Advisor: Jelena Bradic, professor at Cornell University *Ithaca, NY*

- Developed a theoretical framework for unseen-symbol generalization in one-layer transformers by reducing ridge-regularized transformer training to kernel logistic regression in the infinite-width regime.
- Introduced a collision-graph perturbation framework characterizing how finite-sample token collisions affect generalization, and derived probabilistic margin-transfer guarantees for template-based classification tasks.
- Conducted simulation studies validating the role of attention regularization and dataset collision geometry in fresh-symbol generalization.

Inference for High-dimensional Binary Classification with Class Imbalance Jun. 2023 – May 2025
Advisor: Alexander Giessing, now assistant professor at National University of Singapore *Seattle, WA*

- Proposed the bias correction procedure with a convex program based on bias-variance trade-off.
- Proved the consistency and asymptotic efficiency of the estimator in high-dimensional settings and rare events.
- Implmented the algorithm and conducted simulation studies. Our method outperforms the existing inference methods by having smaller variance and better normality under the logistic model with imbalanced sample.
- Currently working on the paper and package, and applying our method on type II diabetes (T2D) data.

Clustered Graphical Modeling for Microbiome Data Oct. 2022 – Mar. 2026
Advisor: Jing Ma, from Fred Hutchinson Cancer Center *Seattle, WA*

- Built a hierarchical Gaussian graphical model with adjacency matrix following a stochastic block model.
- Estimated the precision matrix, model parameters and cluster membership using the variational EM method.
- Implemented the algorithm on the simulated data. It outperformed the existing clustered graphical Lasso algorithms in network discovery and clustering when the network is generated from a stochastic block model.

- Applied our method on S&P 100 and microbiome data, our model achieved higher test accuracy and better interpretation than existing methods.

Activities

Optiver PhD Quant Focus Program Oct. 13–15, 2026
Learn, build, and collaborate on systematic trading during this three-day program. *Chicago, IL*

NeurIPS 2026 Dec. 6–12, 2026
Ongoing poster event. *Sydney, Australia*

Awards

National First Prize | 12^{th} *National College Mathematics Competition (Final)* May. 2021

Finalist | 12^{th} *S.-T. Yau College Student Mathematics Contest – Probability and Statistics* May. 2021

Skills

Languages R (proficient), Python (Numpy, Pandas, Sklearn, Pytorch), LaTeX, C++.

ML/AI PyTorch, scikit-learn, transformers, statistical learning.

Scientific Computing NumPy, Pandas, SciPy.

Research high-dimensional statistics, asymptotic theory, optimization, probabilistic modeling.

Tools Git, Linux, LaTeX, SLURM.